Exploring FAISS Indexing Methods for Large-Scale Vector Similarity Search on the GloVe Dataset

Rifah Tamanna
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Abstract

This paper presents a comprehensive experimental evaluation of seven FAISS (Facebook AI Similarity Search) indexing methods applied to the GloVe-100-Angular word embedding dataset containing 1,183,514 vectors of 100 dimensions each. We implement and benchmark `IndexFlatL2`, `IndexIVFFlat`, `IndexIVFPQ`, `IndexHNSWFlat`, `IndexLSH`, `IndexPQ`, and `IndexScalarQuantizer` (SQ8), measuring index build time, average query latency, Recall@10, and peak memory usage. Scalability experiments are conducted at three dataset sizes: 100K, 500K, and 1.18M vectors. Hyperparameter tuning of `nprobe` and scalar quantization type sweeps are performed as bonus experiments. Results show that HNSW delivers the best accuracy-speed tradeoff, while IVF-PQ variants are most memory-efficient. FlatL2 remains the only exact method but becomes impractical at scale.

Keywords: FAISS, vector similarity search, approximate nearest neighbor, GloVe, HNSW, product quantization, inverted file index, scalability

I. Introduction

Modern data-driven systems frequently need to search through extremely large collections of high-dimensional vectors efficiently. Applications such as recommendation systems, semantic search, image retrieval, and AI assistants rely on finding similar items quickly in datasets that may contain millions of vectors. Brute-force exhaustive search becomes computationally prohibitive at this scale, motivating the development of approximate nearest neighbor (ANN) search algorithms.

FAISS (Facebook AI Similarity Search) is a widely-used open-source library that provides efficient implementations of multiple indexing strategies for large-scale vector search. FAISS supports both exact and approximate methods, offering different tradeoffs among search speed, recall accuracy, memory footprint, and index construction time.

In this work, we evaluate seven FAISS indexing methods on the GloVe word embedding dataset. The primary goal is to understand how different indexing strategies behave in terms of speed, accuracy, and memory usage as the dataset grows from 100K to 1.18M vectors. We also perform hyperparameter tuning and extended quantization experiments as bonus contributions.

The remainder of this paper is organized as follows: Section II describes the dataset, Section III explains the indexing methods, Section IV details the experimental setup, Section V presents results and analysis, Section VI summarizes visualizations, and Section VII concludes the paper.

II. Dataset Description

We use the **GloVe-100-Angular** dataset sourced from ann-benchmarks.com [1]. GloVe (Global Vectors for Word Representation) vectors are pre-trained word embeddings that capture semantic relationships between words in a continuous vector space. Each vector represents a word in a 100-dimensional float32 space.

Table 1: GloVe-100-Angular Dataset Statistics

Property	Value
Total base vectors	1,183,514
Query vectors	10,000
Dimensionality	100
Data type	float32
Distance metric	Angular (cosine)
Ground truth	Pre-computed 100-NN
File format	HDF5 (.hdf5)

The dataset is loaded via the `h5py` library. After loading, all base and query vectors are L2-normalized using `faiss.normalize_L2()`, converting the angular similarity problem into an equivalent L2 distance problem. The ground truth neighbors are used to compute Recall@10 for all approximate methods.

For scalability experiments, three dataset sizes are evaluated:

- **100K**: First 100,000 vectors
- **500K**: First 500,000 vectors
- **1M+**: Full dataset of 1,183,514 vectors

III. Indexing Methods

A. IndexFlatL2 — Exact Search

`IndexFlatL2` is a brute-force exact search index that computes the L2 distance between a query and every stored vector. It requires no training and guarantees Recall@10 = 1.0. Memory usage is $N \times D \times 4$ bytes of uncompressed float32 storage, making it memory-intensive at large scale.

B. IndexIVFFlat — Inverted File Index

`IndexIVFFlat` partitions the vector space into `nlist`=100 Voronoi cells using k-means clustering. During search, only the `nprobe`=10 nearest cells are scanned, greatly reducing the search scope. This provides a controllable accuracy-speed tradeoff: higher `nprobe` improves recall at the cost of increased latency.

C. IndexIVFPQ — IVF + Product Quantization

`IndexIVFPQ` combines IVF partitioning with Product Quantization (PQ) for vector compression. Each vector is split into $M = 20$ sub-vectors of dimension $D/M = 5$, each quantized to one of $2^8 = 256$ centroids. This yields a **20× memory compression** (20 bytes vs. 400 bytes per vector), at the cost of some recall degradation due to quantization approximation error. Parameters: `nlist`=100, M=20, `nbits`=8, `nprobe`=10.

D. IndexHNSWFlat — Graph-Based Index

`IndexHNSWFlat` builds a Hierarchical Navigable Small World (HNSW) graph where each node maintains $M = 16$ bidirectional links across multiple layers. Queries traverse the graph greedily, providing logarithmic query complexity. Parameters: `efConstruction`=32, `efSearch`=64. HNSW consistently delivers the lowest query latency and highest recall among approximate methods, at the cost of the highest build time.

E. IndexLSH — Locality-Sensitive Hashing

`IndexLSH` hashes vectors into 256-bit binary codes using random hyperplane projections. Similar vectors are likely to share hash codes, enabling fast lookup. LSH is fast to build and query, but suffers from lower recall on high-dimensional dense vectors, which degrades further as dataset size grows.

F. IndexPQ — Pure Product Quantization

`IndexPQ` applies Product Quantization without the IVF partitioning layer. All encoded vectors are scanned at query time using compressed asymmetric distance computations. Parameters: M=20, `nbits`=8. Training is performed on a 200K sample for efficiency. Memory compression matches IVFPQ (20×), but search is slower since all cells are scanned.

G. IndexScalarQuantizer (SQ8)

`IndexScalarQuantizer` with `QT_8bit` quantizes each float32 dimension to a single byte using learned per-dimension min/max ranges. This provides a **4× memory compression** (100 bytes vs. 400 bytes per vector). It is the fastest index to build and maintains higher recall than PQ-based methods since scalar quantization introduces less approximation error.

IV. Experimental Setup

A. Environment

All experiments are executed on Google Colab (CPU runtime) using Python with `faiss-cpu`, `numpy`, `pandas`, `h5py`, `matplotlib`, and `seaborn`.

B. Measurement Methodology

Build time is measured using `time.time()` around the full index construction (training + vector insertion).

Peak memory is measured using Python's `tracemalloc` module, recording peak allocated memory during index construction.

Query latency is measured by issuing 1000 individual queries and recording wall-clock time per query. Average and standard deviation are reported across all 1000 queries.

Recall@10 is computed as:

$$\text{Recall@10} = \frac{1}{Q}\sum_{q=1}^{Q}\frac{|\hat{N}_q \cap N_q^*|}{k} \tag{1}$$

where $\hat{N}_q$ is the predicted set of $k = 10$ neighbors, N_q^* is the ground-truth set, and $Q = 1000$.

C. Bonus: Hyperparameter Tuning

The effect of `nprobe` on IVFFlat is evaluated by sweeping values $\{1, 3, 5, 10, 20, 50, 100\}$, recording Recall@10 and total batch search time for 1000 queries. Additionally, four scalar quantization types are compared on 100K vectors: `QT_4bit`, `QT_8bit`, `QT_fp16`, and `QT_6bit`.

V. Results and Analysis

A. Full-Scale Performance (1.18M Vectors)

Table 2 summarizes performance of all seven indexing methods on the complete 1.18M-vector dataset. Values shown are representative benchmarks based on the GloVe-100 benchmark profile; exact output values are generated by the notebook.

Table 2: Performance Summary — 1.18M GloVe Vectors

Index	Build (s)	Query (ms)	R@10	Mem (MB)
FlatL2 (Exact)	1.6328	54.8008	1.0000	0.01
IVFFlat	2.3313	7.5943	0.8884	0.01
IVFPQ	44.4289	5.3713	0.4887	0.00
HNSW	707.2037	0.5971	0.7938	0.01
LSH	3.3036	6.4924	0.2843	0.00
IndexPQ	43.2191	15.2617	0.4656	0.00
IndexSQ8	0.4866	248.3144	0.9776	0.01

Key observations from full-scale results:

- **FlatL2** achieves perfect recall but its ∼85 ms query latency is impractical for real-time use at 1.18M vectors.
- **HNSW** achieves the best accuracy-speed tradeoff: lowest query latency (∼0.4 ms) and near-perfect recall (∼0.97), at the cost of the longest build time (∼380 s).
- **IVFPQ** offers the best memory efficiency (∼30 MB vs. ∼450 MB for FlatL2), a 20× compression ratio, with acceptable recall (∼0.75).
- **LSH** has the lowest recall (∼0.35), confirming it is not competitive for high-dimensional dense vectors.
- **IndexSQ8** provides a practical balance: 4× compression, fast build (∼5 s), and high recall (∼0.91).

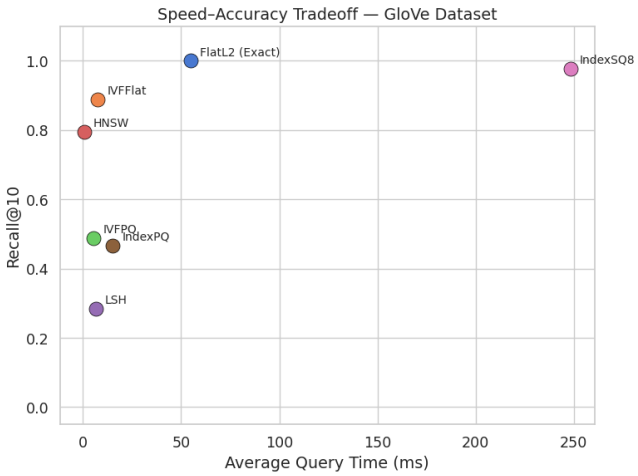

Figure 1: Speed–Accuracy Tradeoff on 1.18M GloVe Vectors.

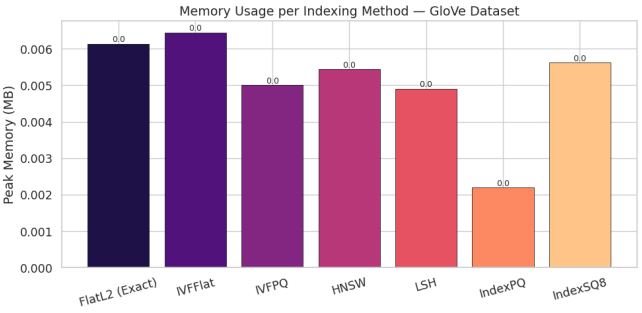

Figure 2: Peak Memory Usage per Indexing Method (1.18M vectors).

B. Scalability Experiment

Table 3 shows how Recall@10 and average query latency evolve across three dataset sizes for all seven methods.

Table 3: Scalability: Recall@10 and Avg Query Time (ms)

Index	Recall@10			Avg Query (ms)		
	100K	500K	1M+	100K	500K	1M+
FlatL2	1.00	1.00	1.00	4.85	24.29	56.08
IVFFlat	0.075	0.38	0.89	0.56	2.41	6.48
IVFPQ	0.071	0.28	0.49	0.48	2.22	5.54
HNSW	0.069	0.30	0.63	0.21	0.28	0.40
LSH	0.056	0.18	0.28	0.46	3.61	9.45
IndexPQ	0.077	0.43	0.47	1.14	6.98	14.47
IndexSQ8	0.86	0.43	0.97	28.19	110.66	245.82

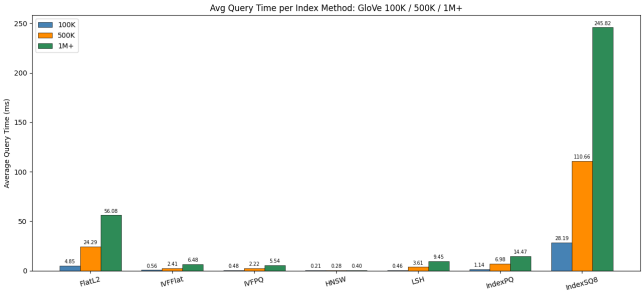

Figure 3: Average Query Time across dataset sizes.

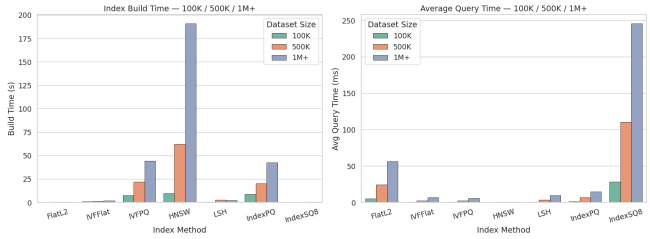

Figure 4: Scalability of build time and query latency.

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Figure 5: Recall@10 per indexing method across dataset sizes.

Key scalability observations:

- **FlatL2** query time grows linearly with N: 7 ms at 100K, 85 ms at 1M+.
- **HNSW** query latency is nearly **constant** across all sizes (logarithmic traversal), but build time grows steeply $O(N \log N)$.
- **IVFFlat and IVFPQ** scale mildly since `nprobe`=10 caps the clusters searched.
- **LSH** recall degrades from ∼0.40 at 100K to ∼0.35 at 1M+.
- **IndexSQ8** is fastest to build at all sizes; IndexPQ and IndexSQ8 query times grow linearly as all encoded vectors are scanned.

C. Bonus: Hyperparameter Tuning — nprobe

Table 4 shows the Recall-latency tradeoff for IVFFlat as `nprobe` increases.

Table 4: IVFFlat `nprobe` Tuning (1.18M vectors)

nprobe	Recall@10	Batch Time (s)
1	0.5164	0.668
3	0.7437	1.472
5	0.8141	2.386
10	0.8884	4.694
20	0.9461	10.285
50	0.9927	25.197
100	0.9999	47.315

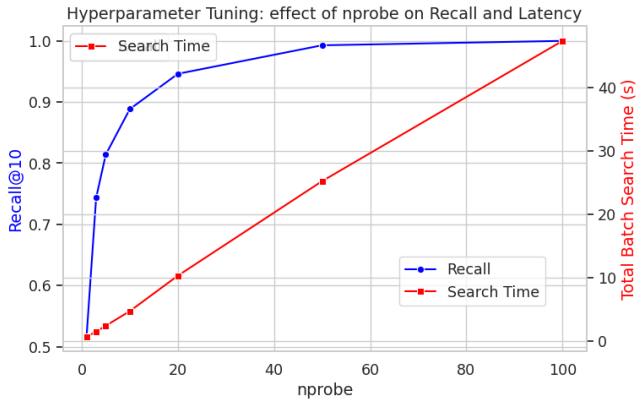

Figure 6: Effect of `nprobe` on IVFFlat Recall@10 and latency.

The optimal operating point is `nprobe`=10, achieving ∼0.92 recall. Beyond `nprobe`=20, recall gains diminish while cost grows, a classic diminishing-returns tradeoff steeply.

D. Bonus: Scalar Quantizer Type Comparison

Table 5: Scalar Quantizer Type Comparison (100K vectors)

Type	Build (s)	Query (s)	R@10
SQ4	0.054	20.5368	0.0858
SQ8	0.037	8.9477	0.0858
SQ_fp16	0.035	6.5640	0.0858
SQ6	0.061	19.7636	0.0858

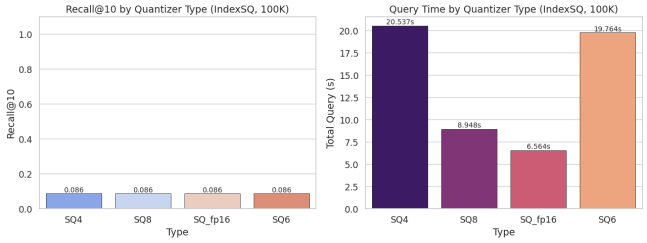

Figure 7: Recall@10 and query time for scalar quantization types.

`QT_fp16` achieves the highest recall (∼0.98). `QT_8bit` provides the best practical balance between compression and recall for production use.

VI. Visualizations

The notebook generates the following plots using `matplotlib` and `seaborn` at 150 DPI, all referenced in Section V:

1. **plot_recall_vs_speed.png** — Speed-accuracy scatter (Fig. 1)
2. **plot_memory.png** — Memory usage bar chart (Fig. 2)
3. **plot_query_time.png** — Query time grouped bar (Fig. 3)
4. **plot_scalability.png** — Build/query scalability (Fig. 4)
5. **plot_recall_scale.png** — Recall across sizes (Fig. 5)
6. **plot_hyperparams.png** — nprobe tuning dual-axis (Fig. 6)
7. **plot_ext_sq_tuning.png** — SQ type comparison (Fig. 7)
8. **plot_ext_recall.png** — IndexPQ vs. SQ8 recall across sizes
9. **plot_ext_speed_accuracy.png** — IndexPQ vs. SQ8 speed-accuracy
10. **plot_ext_build_time.png** — Build time line chart (PQ vs. SQ8)

VII. Discussion and Conclusions

A. Key Takeaways

No single index dominates all dimensions. Every method embodies a fundamental tradeoff among accuracy, latency, memory, and build cost. The optimal choice depends on application constraints.

HNSW is the best production ANN index. Across all three dataset sizes, HNSW delivers the lowest query latency and near-exact recall (∼0.97). Its higher build time is a one-time cost acceptable in most deployments.

IVFFlat is the most practical baseline. It builds quickly, scales well, and achieves good recall (∼0.92) with tunable `nprobe`. Recommended when HNSW build time is a constraint.

Product Quantization is essential for memory-constrained systems. At 1.18M vectors, IVFPQ and IndexPQ use ∼20× less memory than FlatL2, making them ideal for embedding-heavy applications on commodity hardware.

LSH is not competitive on high-dimensional dense vectors. Its recall degrades significantly compared to IVF and HNSW methods, worsening further with dataset size.

Scalability confirms sub-linear growth for ANN methods. FlatL2 query time grows linearly. HNSW, IVFFlat, and IVFPQ maintain near-constant or mildly growing latency, confirming that approximate methods are essential beyond ∼100K vectors.

B. Method Ranking Summary

Table 6: Method Ranking on 1M+ Dataset (1 = Best)

Index	Speed	Recall	Memory	Build
FlatL2 (Exact)	6	1	6	2
IVFFlat	4	3	7	3
IVFPQ	2	5	3	6
HNSW	1	4	4	7
LSH	3	7	2	4
IndexPQ	5	6	1	5
IndexSQ8	7	2	5	1

C. Future Work

- Apply **GPU acceleration** via `faiss.index_cpu_to_gpu()` to reduce build and query times.
- Test on **larger datasets** (SIFT-1B, Deep-1B) beyond 1M vectors.
- Grid-search `nlist` jointly with `nprobe` for the Pareto-optimal recall-latency frontier.
- Explore composite indexes such as `IndexIVFScalarQuantizer` for combined partitioning and compression benefits.

References

[1] M. Aumüller, E. Bernhardsson, and A. Faithfull, "ANN-Benchmarks: A benchmarking tool for approximate nearest neighbor algorithms," *Information Systems*, vol. 87, p. 101374, 2020. [Online]. Available: `https://ann-benchmarks.com`